

Graphs of cube root functions



1 Consider the function $y = \sqrt[3]{x}$.

a Complete the table of values. Round any values to two decimal places if necessary.

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y											

b Sketch the graph of $y = \sqrt[3]{x}$.

c Is $y = \sqrt[3]{x}$ an increasing function or a decreasing function?

d Is there any restriction on the value of x ?

e State the range of the function.

f Does $y = \sqrt[3]{x}$ have a limiting value?

g Does $y = \sqrt[3]{x}$ have an asymptote?

h As x gets larger and larger, what value does y approach?

i For $x \geq 0$, describe the rate of increase of the function as x increases.

j For $x \geq 0$, describe the rate of decrease of the function as x increases.

2 Consider the function $y = -\sqrt[3]{x}$.

a Complete the table of values. Round any values to two decimal places if necessary.

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y											

b Sketch the graph of the function.

c Can the function values ever be positive?

d Can the function value ever be 0?

e State the domain of the function.

f State the range of the function.

g Is $y = -\sqrt[3]{x}$ an increasing function or a decreasing function?

h Describe the rate of increase/decrease of the function $y = -\sqrt[3]{x}$ on either side of $x = 0$.

3 Consider the function $y = \sqrt[3]{-x}$.



a Complete the table of values. Round any values to two decimal places if necessary.

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y											

b Sketch the graph of $y = \sqrt[3]{-x}$.

c Is $y = \sqrt[3]{-x}$ an increasing function or a decreasing function?

d State the domain of the function.

e State the range of the function.

f As x approaches $-\infty$, what does y approach?

g As x approaches ∞ , what does y approach?

4 Consider the function $y = \sqrt[3]{x} + 3$.

a Can y ever be negative?

b As x gets larger and larger, what value does y approach?

c Determine the y -intercept of the curve.

d How many x -intercepts does it have?

e Sketch the graph of $y = \sqrt[3]{x} + 3$.

5 Consider the function $y = -5\sqrt[3]{x}$.

a State the domain of the function.

b State the range of the function.

c Sketch the graph of $y = -5\sqrt[3]{x}$.

6 Consider the function $f(x) = \sqrt[3]{x-1}$.

a Complete following table of values:

x	-7	0	1	2	9
$\sqrt[3]{x-1}$	-2				

b Sketch the graph of the function.

c State the domain of the function.

d State the range of the function.

7 Consider the function $f(x) = \sqrt[3]{x} - 1$.



a Complete following table of values:

x	-8	-1	0	1	8
$\sqrt[3]{x} - 1$	-3				

b Sketch the graph of the function.

c State the domain of the function.

d State the range of the function.

8 Consider the following functions:

i Sketch the graph of the function.

ii State the domain of the function.

iii State the range of the function.

a $f(x) = \frac{1}{2}\sqrt[3]{x}$

b $f(x) = \frac{\sqrt[3]{-x}}{3}$

c $f(x) = \sqrt[3]{\frac{1}{3}x}$

d $f(x) = \sqrt[3]{x-1} + 4$

e $f(x) = \sqrt[3]{-x} - 5$

f $f(x) = \sqrt[3]{\frac{x+4}{2}}$

g $f(x) = \sqrt[3]{-(x-5)}$

h $f(x) = 3\sqrt[3]{x} - 2$

i $f(x) = \frac{\sqrt[3]{4x}}{2}$

j $f(x) = \sqrt[3]{-\frac{x}{2}}$

Characteristics of cube root functions

9 Describe how the function $y = 5\sqrt[3]{x}$ differs from $y = \sqrt[3]{x}$.

10 State the domain and range of $y = \sqrt[3]{x} - 4$.

11 Consider the function $y = \sqrt[3]{x-4}$.

a State the domain of the function.

b State the range of the function.

c Compare the rate of increase of $y = \sqrt[3]{x-4}$ and $y = \sqrt[3]{x}$.